ECA18 – OPTOELECTRONICS AND OPTICAL COMMUNICATION
CO5 – MODEL EXAMINATION

Item	Details
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Course Outcome	CO5 (BTL K4 – Analyze, WK5)
Topics	OTDR, OSA, link budget, eye diagram/BER/Q, ITU-T G.694.1, silicon photonics, splicing, photon counting, coherent detection

Q1. High-Speed Optical Network: Measurement, Link Budget and Sustainability

(i) OTDR and OSA: principles, roles, dynamic range and dead zones (3 marks)

Operating principle of the OTDR. The Optical Time-Domain Reflectometer launches a short, high-power laser pulse into the fibre through a directional coupler. While the pulse travels, **Rayleigh backscatter** (from microscopic density variations) returns continuously, and **Fresnel reflections** return at connectors, breaks and the fibre end. An avalanche photodiode receiver measures the returned power against time, and the time of flight is converted to distance:

$z = c \cdot t / (2n)$ *(n = group index ≈ 1.468; the factor 2 is for the round trip)*

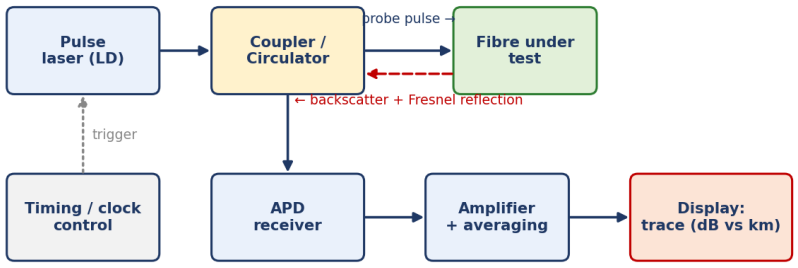


Fig. 1 OTDR block diagram

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The trace plots backscatter level (dB) against distance. The slope of the straight section gives the **fibre attenuation**, $\alpha = [P_1(\text{dB}) - P_2(\text{dB})] / (z_2 - z_1)$ in dB/km. A sudden step shows a localised loss (splice, bend), and a sharp spike shows a reflective event (connector, crack, break). Many pulses are averaged to lift the signal above noise.

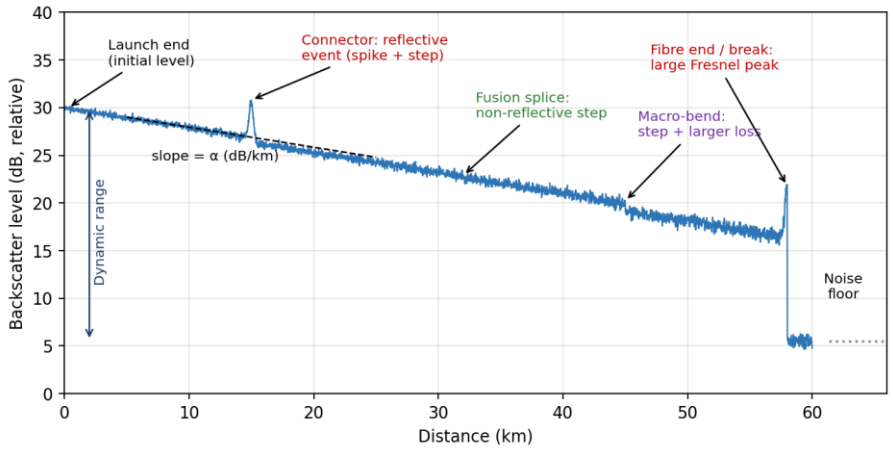


Fig. 2 Typical OTDR trace with events

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Operating principle of the OSA. A grating-based Optical Spectrum Analyzer sends the input light through an entrance slit to a collimator and a rotating diffraction grating, which spreads the wavelengths. Only a narrow band passes through the exit slit to an InGaAs detector, and rotating the grating scans the wavelength. The result is optical power plotted against wavelength.

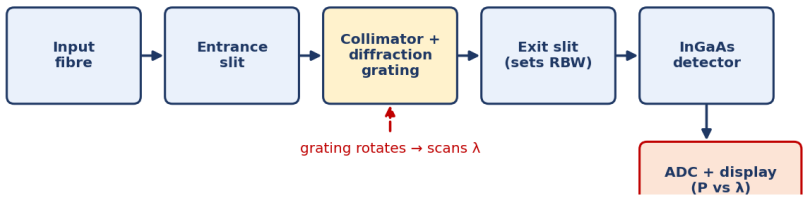


Fig. 3 Grating-based optical spectrum analyzer (OSA)

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Aspect	OTDR	OSA
Domain	Time/distance (power vs distance)	Wavelength (power vs λ)
Main measurements	Fibre attenuation (dB/km), event loss, reflectance, fibre length, fault location	Channel wavelength, channel power, OSNR, SMSR, spectral width, channel crosstalk, amplifier gain and noise figure
Access	One end of the fibre; out-of-service (pulse disturbs traffic)	Tap/monitor port; can be in-service
Key limits	Dynamic range, event and attenuation dead zones, pulse width	Resolution bandwidth (about 0.02–0.1 nm), wavelength accuracy, optical dynamic range (about 60 dB)
Role in this network	Commissioning, fault detection, splice/connector qualification	Checking wavelength allocation, OSNR and compliance with the ITU grid

Dynamic range is the difference (in dB) between the backscatter level at the start of the trace and the RMS noise floor. It sets how far the OTDR can see: the usable length is approximately (DR − 6 dB)/α. For example, a 35 dB dynamic range with α = 0.21 dB/km measures about (35 − 6)/0.21 ≈ 138 km. A longer pulse raises dynamic range but worsens resolution.

Dead zones arise because a strong reflection saturates the receiver for a short time. Dead-zone length ≈ c·W/(2n), where W is the pulse width. The **event dead zone** is the minimum spacing at which two reflective events can be told apart, and the **attenuation dead zone** is the distance after an event needed before attenuation can be measured accurately. Short pulses (for example 10 ns ≈ 1 m) give fine resolution, and long pulses (for example 1 µs ≈ 100 m) give long range. Launch cables are used so that the first connector lies outside the dead zone.

(ii) Link budget, OSNR, eye diagram, BER and Q-factor (3 marks)

Optical link budget. The power budget requires that the transmitted power minus all losses and penalties is still above the receiver sensitivity:

$$P_{tx} - P_{rx,min} = \alpha \cdot L + N_c \cdot \alpha_c + N_s \cdot \alpha_s + \text{dispersion penalty} + \text{system margin}$$

Item	Value (example)	Remark
Transmit power	0 dBm	Laser launch power (limited by nonlinearity and eye safety)
Fibre loss, 80 km × 0.22 dB/km	17.6 dB	Attenuation at 1550 nm
Connectors, 2 × 0.5 dB	1.0 dB	Each mated pair adds insertion loss
Splices, 10 × 0.1 dB	1.0 dB	Fusion splices about every 8 km
Dispersion penalty	1.0 dB	Power needed to compensate pulse broadening
System margin	3.0 dB	Ageing, repairs, temperature, connector re-mating
Total loss	**23.6 dB**	Received power = 0 − 23.6 = −23.6 dBm
Receiver sensitivity	−28 dBm	Required for BER 10^{-9} (Q = 6)
Spare margin	**4.4 dB**	Link is feasible

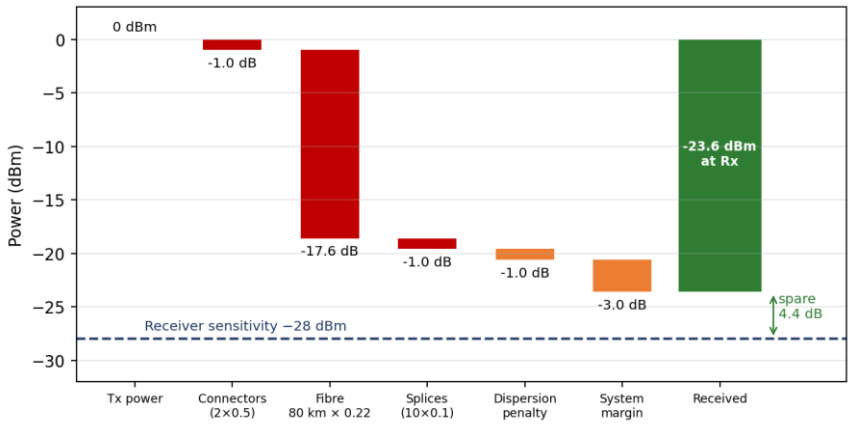


Fig. 4 Optical link power budget (worked example)

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- **Power margin:** extra dB reserved for ageing, repair splices, temperature drift and component tolerances (typically 3–6 dB).
- **Dispersion penalty:** chromatic dispersion (D ≈ 17 ps/nm/km) and PMD broaden pulses and cause inter-symbol interference, which closes the eye. A common rule is that broadening should remain below about 0.1–0.25 of the bit period, which limits a 10 Gb/s link to roughly 60–80 km without compensation, and the receiver needs 1–2 dB more power.
- **OSNR:** in an amplified link, each EDFA adds ASE noise. For N identical spans, OSNR(dB) ≈ 58 + P_ch(dBm) − NF − Span loss − $10 \cdot \log_{10} N$ (0.1 nm reference bandwidth). Example: 58 + 0 − 5 − 20 − 7 = **26 dB** for 5 spans (NF = 5 dB, span loss = 20 dB).

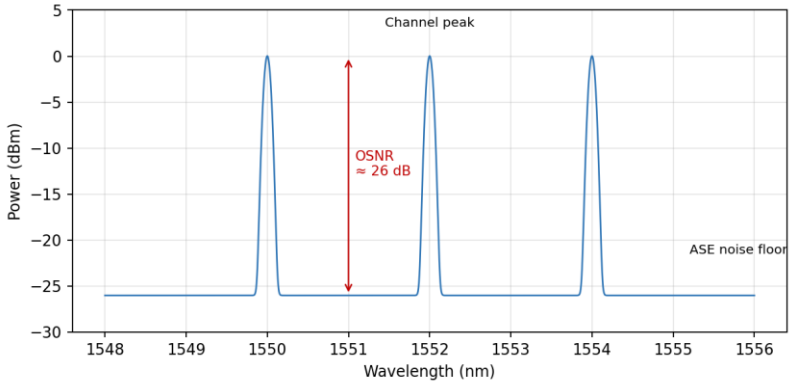


Fig. 5 OSA display: 3 WDM channels, ASE floor and OSNR

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Eye diagram, BER and Q-factor. An eye diagram overlays all bit intervals of the received signal on an oscilloscope. A wide-open eye means little ISI, noise and jitter. **Eye height** shows noise margin, **eye width** shows timing margin, and the crossing spread shows jitter.

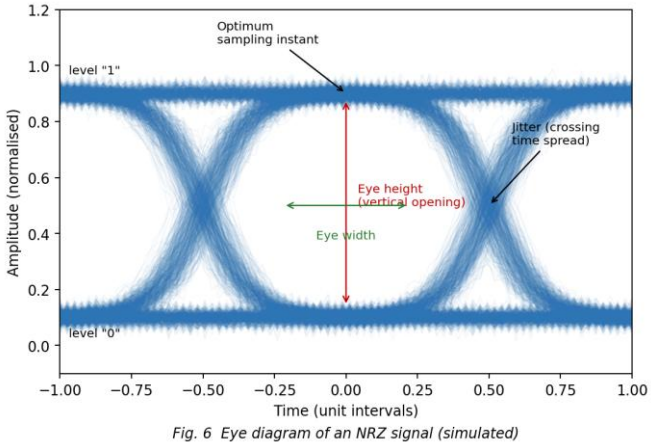


Fig. 6 Eye diagram of an NRZ signal (simulated)

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The **Q-factor** quantifies the eye, and the **BER** follows from it for Gaussian noise:

$Q = (\mu_1 - \mu_0) / (\sigma_1 + \sigma_0)$ $\quad BER = \tfrac{1}{2} \cdot erfc(Q/\sqrt{2})$ $\quad Q(dB) = 20 \cdot \log_{10} Q$

Q (linear)	Q (dB)	BER	Meaning
3	9.5	1.3×10^{-3}	Poor; needs FEC
4	12.0	3.2×10^{-5}	Marginal

Q (linear)	Q (dB)	BER	Meaning
6	15.6	1.0×10^{-9}	Standard target (no FEC)
7	16.9	1.3×10^{-12}	Excellent; high-quality link

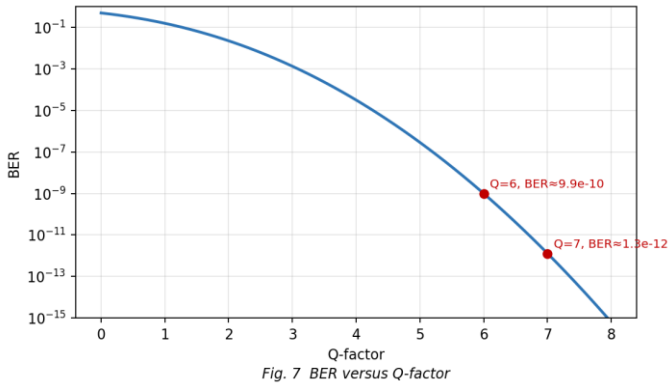


Fig. 7 BER versus Q-factor

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In practice, a measured Q (or BER) below target tells the engineer to increase power, reduce dispersion, lower noise or add forward error correction.

(iii) ITU-T G.694.1 grid, channel spacing and Nyquist spacing (2 marks)

ITU-T G.694.1 defines the DWDM frequency grid, anchored at **193.1 THz** ($\lambda \approx 1552.52$ nm) with spacings of 100, 50, 25 and 12.5 GHz, and a flexible grid with 12.5 GHz granularity (slot width = n × 12.5 GHz). Its significance:

- **Interoperability:** transmitters, mux/demux filters, ROADMs and amplifiers from different vendors line up on the same channel frequencies.
- **Predictable crosstalk and planning:** fixed channel centres make channel allocation, monitoring (OSA) and upgrades simpler, and the laser wavelength-locking tolerance can be specified.
- **Capacity planning:** the C-band (about 4.8 THz) holds 48 channels at 100 GHz or 96 channels at 50 GHz.

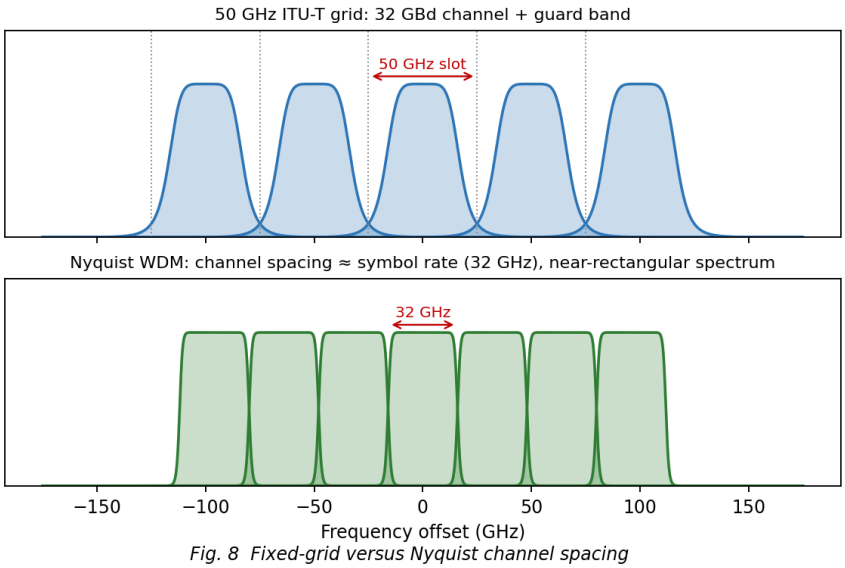


Fig. 8 Fixed-grid versus Nyquist channel spacing

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Channel spacing and spectral efficiency (SE = bit rate/spacing). Narrower spacing packs more channels into the same fibre, but increases inter-channel crosstalk and four-wave mixing, and demands tight laser and filter stability. **Nyquist WDM** shapes each channel with a root-raised-cosine or sinc pulse so that its spectrum is nearly rectangular; the channel spacing can then equal the symbol rate ($\Delta f \approx R_s$) with almost no ISI or guard band.

Scheme	Spacing	Bit rate	Spectral efficiency
100 Gb/s DP-QPSK on 50 GHz grid	50 GHz	100 Gb/s	2.0 b/s/Hz
Same signal, Nyquist spacing	≈ 33 GHz	100 Gb/s	≈ 3.0 b/s/Hz
Flexible grid with 16QAM (example)	≈ 37.5 GHz	200 Gb/s	≈ 5.3 b/s/Hz

(iv) Energy-efficient transceivers, optical interconnects and silicon photonics (2 marks)

Data traffic grows faster than energy budgets, so network power consumption and heat (not just speed) become the limit. Energy per bit is therefore a key figure of merit (pJ/bit).

- **Energy-efficient transceivers:** lower laser drive current, efficient DSP/driver ICs, integrated lasers and modulators, and advanced modulation with FEC all reduce pJ/bit. A 400G pluggable using about 12–15 W corresponds to roughly 30–40 pJ/bit, and the target for next-generation links is below 5–10 pJ/bit.
- **Optical interconnects:** fibre or waveguide links replace copper for rack-to-rack, board-to-board and chip-to-chip links, giving higher bandwidth density, lower loss at high speed, no crosstalk or EMI, and lower power per bit over distance.
- **Silicon photonics:** modulators (Mach–Zehnder or ring), Ge photodetectors, waveguides and multiplexers are fabricated on silicon with CMOS foundry processes, giving low cost, high volume, compact size and integration with electronics (co-packaged optics).

Aspect	Electrical (copper)	Optical / silicon photonics
Bandwidth–distance	Falls quickly above 25–50 Gb/s	Very high, nearly distance-independent
Energy per bit	Rises with speed and distance	Low and nearly flat
EMI / crosstalk	Significant	Immune
Size / cabling	Bulky, heavy	Thin, light, dense (WDM on one fibre)
Cost trend	Cheap at short reach	Cost falls with CMOS volume

Together these measures give sustainable, high-performance networks: lower electricity and cooling cost, higher port density and scalable bandwidth.

Q2. Long-Distance Fibre Network: Instruments, Joints, Receiver Limits

(i) Power meters, dispersion and PMD analyzers; splicing and connectors (3 marks)

Instrument	Working principle	Application in link evaluation
Optical power meter	A calibrated Ge or InGaAs photodiode converts optical power to current; the reading is shown in dBm at the selected wavelength	Measures transmitter output and received power; with a stabilised source, insertion loss = P_ref − P_meas (end-to-end loss test and budget verification)
Chromatic dispersion analyzer	Phase-shift or time-of-flight method: a tunable laser is modulated, and the group delay $\tau(\lambda)$ is measured at each wavelength; $D = (1/L)\cdot d\tau/d\lambda$	Finds D (ps/nm/km) and zero-dispersion wavelength, which sets the dispersion penalty and the length of dispersion compensation needed
PMD analyzer	Interferometric (INTY) or Jones-matrix eigenanalysis measures the differential group delay (DGD) between the two polarisation modes	Gives PMD in ps (coefficient in ps/√km). Rule of thumb: mean DGD < 0.1 × bit period (e.g. < 10 ps at 10 Gb/s). Critical for ≥10 Gb/s links

Joining techniques and their influence. Loss at a joint comes from lateral offset, angular tilt, end gap, core-diameter/NA mismatch, Fresnel reflection and contamination.

Joint type	Insertion loss	Return loss	Reliability / use
Fusion splice	0.02–0.1 dB	> 60 dB	Permanent, strong, best for long-haul; needs a splicer and protective sleeve
Mechanical splice	0.1–0.5 dB	30–40 dB (index gel)	Quick and reusable, used for emergency repair; higher loss and temperature sensitivity
Connector, PC	0.2–0.5 dB	30–40 dB	Demountable; physical contact; lower reflection performance
Connector, UPC	0.2–0.3 dB	> 50 dB	Polished finish; standard for digital links
Connector, APC (8° angle)	0.2–0.3 dB	> 60 dB	Reflection is deflected into the cladding; for analogue video, PON and high-power links
Connector families	FC, SC, LC, ST, MPO	—	LC for high-density transceivers, SC for PON/FTTH, MPO for multi-fibre data-centre links

Each additional joint adds loss and potential reflection, so fusion splicing is used for the long fixed route and connectors only at patch panels and equipment. Dirty end-faces are the most common cause of failures, and cleaning plus inspection ensures reliability.

(ii) OTDR fault identification, dynamic range and dead zones (3 marks)

The OTDR identifies a fault by the shape of its trace at the corresponding distance $z = c \cdot t/(2n)$:

Fault / event	Trace signature	Interpretation
Fusion splice	Small step down (0.02–0.1 dB), no spike	Non-reflective loss; high step means a poor splice
Connector / mechanical splice	Spike plus a step down	Reflective event; reflectance and loss both measured
Macro-bend	Step down with no spike; larger at 1625 nm than at 1550 nm	Bending-induced loss (use two wavelengths to confirm)
Crack / break	Large reflection followed by drop to noise (open end)	Location of break gives the distance to repair
Fibre end	Large Fresnel spike (about 4 %) then noise	Total length of the link

Significance of dynamic range. It limits how far the trace can be observed above noise. A higher DR allows the far end and distant events to be seen with accurate loss values, and averaging time can be traded for DR (each doubling of averaging time gains about 0.75 dB). For 35 dB DR and 0.21 dB/km, the measurable length is about 138 km.

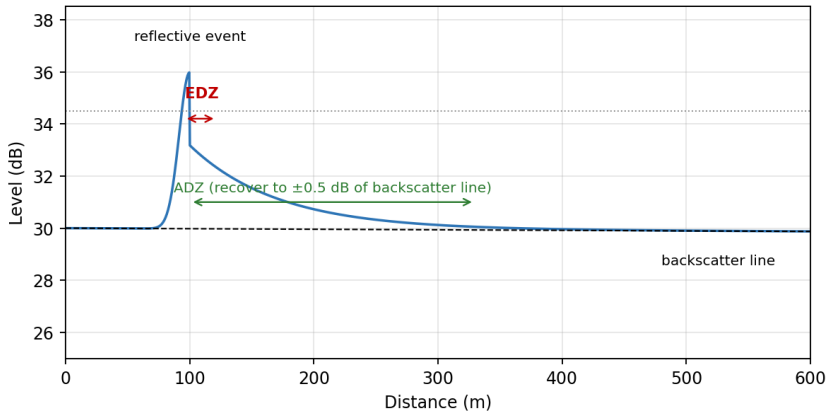


Fig. 9 Event dead zone (EDZ) and attenuation dead zone (ADZ) after a reflective event

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Parameter	Definition	Significance
Event dead zone (EDZ)	Minimum distance between two reflective events that can still be resolved separately (measured 1.5 dB below the peak)	Sets the ability to locate closely spaced connectors, for example in patch panels
Attenuation dead zone (ADZ)	Distance after a reflective event before the trace returns within about 0.5 dB of the backscatter line	Sets how soon loss and attenuation can be measured accurately after an event

Parameter	Definition	Significance
Relation to pulse width	Dead zone ≈ c·W/(2n): 100 ns → ≈ 10 m; 1 μs → ≈ 100 m	Short pulse: better resolution; long pulse: higher DR. ADZ is always longer than EDZ

(iii) Extinction ratio, jitter and Q-factor in link diagnostics (2 marks)

Parameter	Definition	Effect on signal quality	Effect on BER
Extinction ratio (ER)	ER = $10\cdot\log_{10}(P_1/P_0)$ dB; typical 8–13 dB	Low ER leaves optical power in the "0" level, which reduces eye height	Power penalty $\delta = 10\cdot\log_{10}[(r+1)/(r-1)]$, $r = P_1/P_0$: ER 10 dB → 0.87 dB; ER 6 dB → 2.2 dB
Jitter	Timing variation of signal edges (ps RMS or UI), caused by ISI, noise, clock instability	Narrows the horizontal eye opening and sampling margin	Sampling away from the optimum raises errors; excessive jitter gives errors even at high SNR
Q-factor	$Q = (\mu_1 - \mu_0)/(\sigma_1 + \sigma_0)$	Summarises the vertical eye opening relative to noise	BER = ½·erfc(Q/√2): Q = 6 → 10^{-9}, Q = 7 → 10^{-12}

Relationship. A low ER reduces $\mu_1 - \mu_0$, jitter shrinks the effective sampling window, and noise raises σ. All three lower Q and therefore raise BER. In diagnostics, a measured eye with low ER points to the transmitter bias or modulator, a wide crossing band points to jitter or ISI (dispersion, bandwidth), and a thick "1" level points to noise or ASE. Q is used to compare these effects in one number.

(iv) Photon counting, shot-noise-limited detection, homodyne and heterodyne (2 marks)

Photon counting. At very low power the detector (SPAD, photomultiplier or superconducting nanowire detector) registers individual photons. The number of photons per bit interval follows Poisson statistics, $P(n) = e^{(-\bar{N})}\cdot\bar{N}^n/n!$. For ideal on–off keying with no dark counts, an error occurs only if no photon arrives for a "1" bit, giving $BER = \frac{1}{2}\cdot e^{(-\bar{N})}$. BER = 10^{-9} needs $\bar{N} \approx 20$ photons per bit; this is the **quantum limit** of direct detection.

Shot-noise-limited detection. The detector is shot-noise limited when the (amplified) signal or local-oscillator shot noise exceeds thermal and amplifier noise. In a coherent receiver a strong LO power raises the signal current ($\propto \sqrt{(P_s\cdot P_{LO})}$) so that shot noise from the LO dominates, $SNR \approx \eta\cdot P_s/(h\nu\cdot B)$, and the receiver reaches the quantum limit.

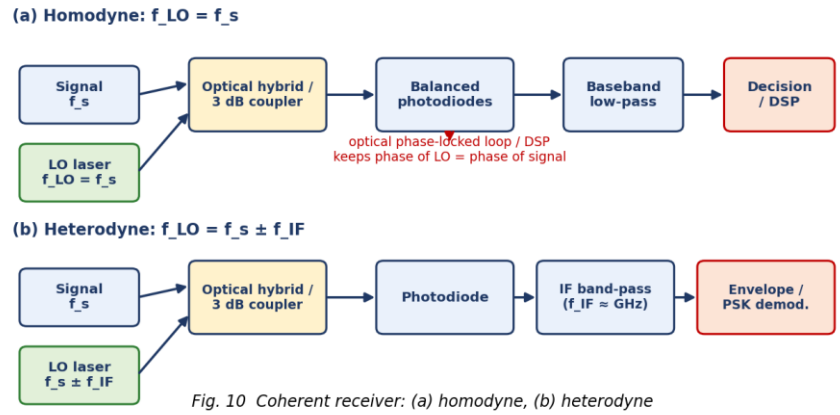


Fig. 10 Coherent receiver: (a) homodyne, (b) heterodyne

Feature	Homodyne detection	Heterodyne detection
LO frequency	Equal to the signal frequency (f_LO = f_s)	Offset by an intermediate frequency (f_LO = f_s ± f_IF)
Output	Baseband electrical signal directly	Electrical signal at f_IF, then demodulated to baseband
Phase requirement	Needs phase locking (optical PLL) or DSP phase recovery	Phase need not be locked; the IF carrier is recovered electrically
Sensitivity (PSK)	Best: about 9 photons/bit for BER 10^{-9}	3 dB worse: about 18 photons/bit
Receiver bandwidth	Lower (baseband)	About twice as high (IF band)
Complexity	Higher optical complexity; supports I/Q detection	Simpler locking, higher electronic bandwidth
Applications	Coherent QPSK/QAM receivers (with DSP), optical phase measurement, quantum communication	Coherent systems with IF, FMCW LiDAR/Doppler sensing, spectroscopy and laser linewidth measurement

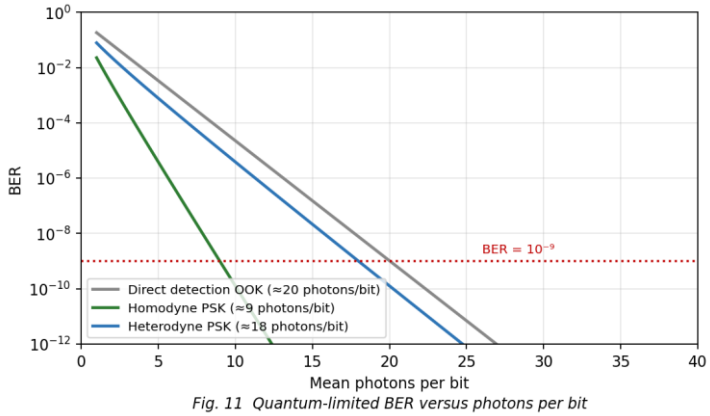


Fig. 11 Quantum-limited BER versus photons per bit

Coherent detection gives about 3–20 dB better sensitivity than direct detection, supports phase and polarisation multiplexing, and uses digital signal processing to compensate dispersion and PMD, which is why modern long-haul systems use intradyne (near-homodyne) receivers.